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**AND FIRST INSPECTION**

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## **SUBSECTION 4.9 RELIABILITY RUNS, PERFORMANCE TESTING AND FIRST INSPECTION**

### **4.9.1. RELIABILITY RUNS**

For the Add-On blocks MT2, MT3 and MT4 the reliability runs shall be performed for the whole respective Add On Block after the Bidder has notified the Owner that the Block is ready for commercial service.

Provided the Owner is satisfied that the Plant is ready for commercial service, the Add On Plant with all its auxiliaries is required to operate without failure or interruption of any kind for a period of 30 days follow the instruction start stop operation from Java Dispatch Center (P3B). In this period the Plant shall operate under the various modes of operation as stipulated by the Owner and under the working conditions of the station within the limits of output specified, either continuously or intermittently, as may be more convenient, for the Owner.

The plant will be operated by the station staff under the supervision of the Bidder during the reliability test period.

The Bidder shall be allowed to make any minor adjustments which may be necessary, provided that such adjustments do not in any way interfere with or prevent the commercial use of the plant by the Owner, or result in reducing the output or decreasing the efficiency.

Should any failure or interruption occur in any portion of the Add On Plant operation or failure start stop operation due to, or arising from, faulty design, materials, workmanship or omissions, incorrect erection, incorrect or inadequate instruction or supervision by the Bidder's supervisors, sufficient to prevent safe and full commercial use of plant, the reliability run shall be considered void and the reliability test period of 30 days shall recommence after the Bidder has remedied the cause of defect.

All adjustments made by the Bidder shall be recorded by him to the satisfaction of the Owner.

Within one month of the completion of the reliability run the Bidder shall submit a report covering the observations and adjustments made during the period.

In respect of this section, the Bidder shall take responsibility for any sub-Bidder.

As soon as practicable before the 30 day's reliability run the guarantee performance tests shall be carried out on the plant.

A Taking Over Certificate shall be issued after the completion of a satisfactory Reliability Run and Performance Test. The requirements of the General Conditions of Contract shall apply.

#### **4.9.2. GUARANTEE PERFORMANCE TESTS**

##### **4.9.2.1 General**

Guarantee performance tests, which may be a repeat in part of test carried out at the manufacturer's works, and any other tests the owner may require to determine that the plant is in accordance with the Specification and Guarantees shall be carried out on the gas turbine generators in open cycle and the combined cycle block as soon as possible by the Bidder, to the requirements of the Owner.

Load, fuel and electricity shall be provided by the Owner except where otherwise specified.

The Bidder shall co-operate with the Owner by organizing and setting up the temporary utilization of unassigned inputs of the Data Acquisition and Display System, and by the production of application programs for temporary logs and displays.

The Bidder shall submit to the Owner, not later than 6 months before the tests are to be carried out, a proposed procedure for the guarantee performance tests. This procedure shall indicate the date and programme anticipated for the test, the arrangement and form of the test, log sheets, duration, control periods, readings to be taken and numbers of observers required as previously agreed with the Owner.

The instruments used for these tests may be those provided with the plant; additional precision grade test instruments and signal sources shall be supplied by the Bidder to meet the accuracy requirements for carrying out the various tests under this Section, where necessary, to the satisfaction of the Owner. Tolerance for instrument uncertainty shall not be allowed.

Additional test instrumentation shall also be provided by the Bidder to enable spot checks to be made during the tests to ascertain in the accuracy of the plant instruments.

The additional test instruments used for the tests shall remain the property of the Bidder.

A certified calibration curve for each test instrument shall be provided and the Bidder shall include for the provision of any recalibration certificates which may be required by the Owner.

All flow meters orifice plates, venturi meters, fuel flow meters and thermocouple with their indicators shall be certified by an approved independent authority.

Should the plant or any portion thereof fail under these tests to give the performance required, then any further tests which may be considered necessary by the Owner shall be carried out in a similar manner, but the whole cost of the repeated tests shall be borne by the Bidder, including fuel, electricity, water and other consumables necessary to carry out the tests.

In the event that the Guarantee Performance Test is delayed due to reasons of the Bidder's responsibility, the Bidder shall not be allowed any factor of aging allowance. In the event that this test is delayed due to reasons outside the responsibility of the Bidder then the Bidder will be allowed aging allowance. The Bidder then shall state in his proposal the percentage ageing allowance to apply to the guaranteed power output.

#### 4.9.2.2 Performance Guarantees

In addition to complying with the requirements of the Specification, the Bidder shall meet the specific guarantees of performance given in the Schedules of Guaranteed Performance.

Performance tests to be carried out **after** the Reliability Run for the following:

- a. Block 2, Block 3 and Block 4 Add-On Steam Turbine Performance, **Auxiliary power consumption** and Complete Combined Cycle (as described on Schedule 6, Plant Performance Guarantee)

ADD1  
-IIH  
-001

#### 4.9.2.3 Tolerance on Guarantee Figures

There shall be no tolerance applied to the fully corrected measured values for purposes of comparison with the guaranteed values.

The guaranteed figures given in the Schedule-6 shall be deemed to include all tolerances.

The above shall apply notwithstanding anything to the contrary stated in the approved or any other standard or code covering performance tests. The Bidder is requested to state in his Proposal the measuring tolerances included in his Power Output Guarantees.

#### 4.9.2.4 Guarantee Test Procedure

During tests, the Plant shall be operated by the Owner's staff under the direction and supervision of the Bidder's representative.

The results of tests on Site shall be correlated to the specified conditions by pre-determined correction factors, which shall be defined by means of correction curves mutually agreed between the Bidder and the Owner. These curves shall be reproduced individually in graphical form and bound in with the Operating and Maintenance Instructions.

The curves will be utilized on Site to determine corrected performance figures which will be compared to the guarantees for the plant being tested.

The Bidder shall include a list of all instruments he shall provide for the purpose of the Performance Tests for the plant on the relevant Schedule in Part 5 together with details of their type, range and accuracy conforming to the requirements of ASME - PTC 46 and also include showing the measuring point for the performance test. The drawing shall be submitted and attached herein.

The power absorbed by the individual auxiliaries where relevant in the course of all tests shall be accurately measured and recorded.

Performance test Add on Combined Cycle shall be refer to ASME PTC 6.2, 2011 and ASME PTC 4.4

The values used in performance calculations shall be the arithmetical average of the observations made, and recorded during the tests. A minimum of twenty one observations shall be taken during each test at regular intervals not exceeding three minutes.

For one hour proceeding the test period, the load and all other conditions shall be in accordance with the test requirements and all readings shall be taken during this hour.

During the test period the conditions shall be held as steady as possible compatible with safe and effective operation of the unit. The heat recovery steam generator shall not be blown down and the fuel condition should be similar at the start and finish of the test period.

The test load and operating conditions shall be maintained for one hour after the termination of the test period, if needed to verify the quality of the data measured during the test, and essential readings shall be taken to show that this requirement has been complied with.

### **Measurements**

- (1) The anticipated points of measurement together with necessary isolation during the tests shall be indicated on Proposal Drawings and on the Test Procedures. The Bidder shall indicate in the Schedules in his proposal the description of the means of measurement of the necessary parameters together with the anticipated standard and accuracy of the instrumentation.
- (2) Suitable current and voltage transformers shall be provided in the contract for the measurement of auxiliary power.
- (3) The Bidder shall provide precision grade watt meters for the measurement of electrical output and auxiliary power consumption. These instruments shall comply with the Technical Requirements.
- (4) The voltage and current transformers associated with the kilowatt-hour meters shall be provided under this Contract and shall comply with the requirements of the test standards.
- (5) Fuel flow measurement shall be to an approved standard. The measurement device shall have accuracy equal or better than 0.5%.

- (6) Calorific value measurement shall be to an approved standard and to accuracy better than 0.35%.

#### **4.9.2.5 Add-On Block Guarantee Tests**

##### **4.9.2.5.1 General**

Add-On Block guarantee tests shall be conducted before the Common Reliability Run for the complete Add-On Power Plant.

The Bidder may in this period make minor adjustments to the HRSG and steam turbine generator to ensure they are suitable for testing.

Tests shall be carried out on the overall combined cycle block on both, natural gas and HSD oil fuels, separately in accordance with the following requirements:

- (1) If the Bidder so desires, the plant will be run for a reasonable time at any load up to that at which the test is to be made. A request for this run, starting duration and load, must be made in writing to the Owner three days before the official test. There shall be a suitable stabilization period before the commencement of each test run when a change of plant load has taken place. Prior to the test the Bidder is at liberty to carry out one compressor/turbine cleaning cycles.
- (2) The generator output shall be determined in accordance with the appropriate standard to suit the existing ambient conditions.

Heat consumption tests shall be carried out at load as defined in Part 5, Schedule 6 and shall be run for a period of one hour.

- (3) Samples of the flue gas used during the tests shall be taken at the beginning, during and on completion of the guarantee performance tests. The necessary apparatus for taking and containing the flue gas samples shall be supplied by the Bidder under this Contract. Three separate samples shall be taken at each of the three sampling period making nine samples in all. One of each set of three samples shall be fully analyzed by an independent authority and the costs of three analyses shall be borne by the Bidder. The other two samples of each set of three shall be properly labeled and one handed over to the Owner and one kept as spare.

The characteristic to be determined shall refer to sampling flue gas in Bid Document Book **II B part 4 Section 3 M-301 Heat Recovery Steam Generator and Auxiliary Equipment**.

- (4) The term full load shall mean the maximum load which is obtainable at the Site ambient conditions prevailing at the time of the tests with the gas turbine running at its designed mean maximum turbine inlet temperatures

The Bidder shall include in the Schedule of Prices a price for the deletion of the guarantee performance tests. The Owner reserves the right to delete one or more tests.

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-002

#### **4.9.2.5.2 Specific Requirements**

After completion of the guarantee performance tests on the Combined Cycle Block the Bidder shall provide within thirty days full performance calculations for each test, according to Part 5, Schedule 6.

The data in the form of calculation sheets shall conform to the format approved by the Owner and shall clearly identify the values measured in the test and shall include corrections for the effects on deviations from design at the time of the tests of:

- (1) Ambient air temperature and pressure
- (2) Humidity
- (3) Generator power factor and frequency
- (4) Turbine generator speed
- (5) Feedwater temperature from the deaerator
- (6) Cooling water inlet temperature (to condenser and auxiliaries)
- (7) Cooling water inlet flow (to condenser and auxiliaries), if applicable
- (8) Existing Gas Turbine exhaust flow conditions

Necessary data shall be logged to ensure that requirements stated above can be fully complied.

The Combined Cycle/ Add-On Block tests shall be conducted with:

- (1) The zero make-up water supply to the condenser isolated.
- (2) The CW sea water system unitized throughout by isolating the CW pump and dedicating them to the Block under test.

Steam plant auxiliaries running for the Combined Cycle Block limited to equipment defined in Schedule-6.

Should any plant or any portion thereof fail under the test to give the performance required, any further tests which may be considered necessary by the Owner will be carried out by the Owner, but the whole cost of the repeated tests shall be borne, by the Bidder.

#### **4.9.2.5.3 Correction Curves**

The Bidder is to submit with his proposal curves giving the corrections to the performance of the steam turbines for variations from the specified conditions at the design output as may be applicable to the Plant. These curves will allow test results calculated at the Site to be corrected back to the design conditions to judge if the test completed is satisfactory.

Each of these curves is to have a range to suit the specified conditions. Curves shall also be submitted to cover operation of the Combined Cycle Block at

guaranteed loads, as per Part 5, Schedule 6. The curves are to show, in particular, the following relationship:

- (1) Variation in steam turbine power with CW inlet temperature and flow.
- (2) Change in steam turbine generator power output and efficiency with power factor, and frequency (speed).
- (3) Change in steam turbine generator power output with variation of feedwater outlet temperature.
- (4) Change in steam turbine generator power output with variation of gas turbine exhaust flow conditions

If the Bidder considers that the above list is inadequate to meet the performance calculation requirements he shall provide whatever other curves are necessary.

#### **4.9.2.5.4 Rejection of Plant**

Rejection of Plant is dealt with in Part 2, General Conditions of Contract, Clause 38.

#### **4.9.2.5.5 Test Results**

Prior to the running of the performance tests, agreement should be reached between the parties concerning the fundamental data obtained from the test. The Bidder shall submit to the Owner, (six) 6 months prior to the test, the method by which he proposes to calculate the results based on the measured data.

- (1) The result of such test shall be correlated to the specified design guarantee condition

There shall be only the tolerance stated in Section clause 4.9.2.3 of this section applied to the fully corrected measured heat consumption for purposes of comparison with the guarantees.

In the determination of fuel properties, the analysis shall be made by independent laboratory mutually agreeable to the Parties, and this analysis shall then be used in computing the Block efficiency.

A full and completed report of all the performance tests is to be produced and six (6) copies issued to the Owner within one month of the completion of each test on Site. The official performance test reports shall include, as a minimum, the following details:

- (1) Full adapted procedure approved by the Owner.
- (2) Instrumentation details and calibrations.
- (3) Full schematic of the air, fuel, flue, gas, steam and feed system together with instrument locations.



- (4) The standard to which the test was carried out and the Code of Practice followed plus the steam tables and other reference data used.
- (5) The operating conditions prior to the tests.
- (6) Variation in system frequency/turbine speed.
- (7) Summary of test readings.
- (8) Results of the performance test.

#### **4.9.2.5.6 Test Instrumentation**

All test instrumentation for the performance and efficiency tests shall be supplied by the Bidder and shall be retained until the satisfactory conclusion of all such tests at Site. All costs associated with the supply, calibration, installation and return of the test instrumentation shall be included in the contract, identified as indicated in the Schedule of Prices. A list of test instrumentation to be provided shall be included in the Technical Schedules. Such instrumentation shall be to National Physical Laboratory Standard or other equivalent standard to the approval of the Owner. Calibration procedure and standard are also subject to the approval of the Owner. Batch calibrations shall not be accepted.

In the event that the Owner wishes to purchase any of the above instrumentation subsequent to the performance tests, the Bidder shall indicate in the appropriate schedule in the Schedule the optional sums payable over the Contract Price for each item.

#### **4.9.2.6 Heat Recovery Steam Generator (HRSG)**

##### **4.9.2.6.1 Steam Purity Test**

A steam purity test shall be performed on the HRSG. On line monitoring shall include, conductivity using permanently installed fast samplers, especially during load changes or unstable conditions when mechanical carryover is more likely. Silica concentration shall be performed by manual analysis.

##### **4.9.2.6.2 Operating test**

The entire systems shall be tested to ensure that it operates satisfactorily at the normal voltage.

#### **4.9.3. INSPECTION AT END OF GUARANTEE PERIOD**

##### **4.9.3.1 Add-On Block**

A visual outside inspection, shall be carried out to the Add-On Block component at the end of the guarantee period and before the release of retention monies for the Combined Cycle Block. The Plant to be inspected shall include, but not be limited to steam turbine generator, condenser, HRSG and drums, pumps, deaerator and other equipment to be agreed by the Owner.

**4.9.3.2 Common Auxiliary Plant**

Inspection shall also be performed to all Common Auxiliary Plant such as Desalination, Water Treatment, Chlorination, etc.

**4.9.3.3 Inspection Procedures**

Inspection procedures shall be submitted to the Owner for review and approval not later than two (2) months before the scheduled inspection. The inspection shall be conducted under conditions mutually agreed by both parties.

(Last page of Subsection 4.9)